

- Battery (VBAT)
- Always on 1.8V up to 750mA (1V8)
- System voltage between VBAT and VUSB (Vsys)
- Three software-controlled digital supplies (VD0, VD1, VD2)
- Two software-controlled analog supplies (VA0, VA1)

- Default
- VD0: Off
- VD1: 1.8V or higher (Buck-Boost)
- VD2: Off
- VA0: Off
- VA1: 1.5 or 3.0V depending on config
- 1V8: Always On

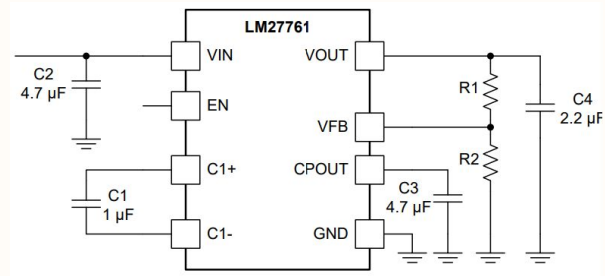
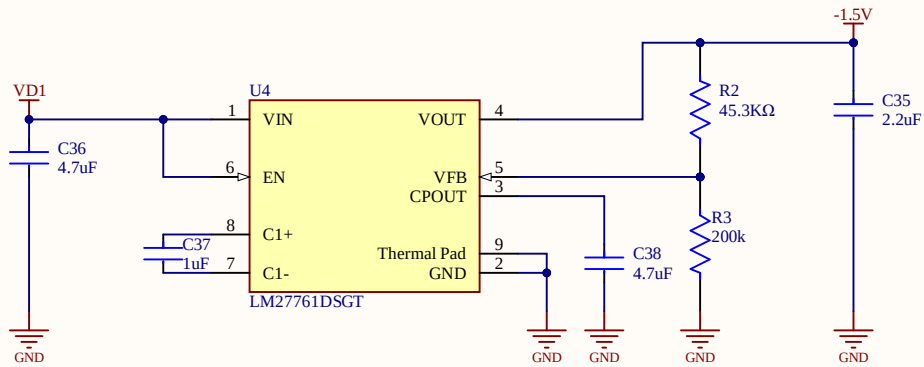
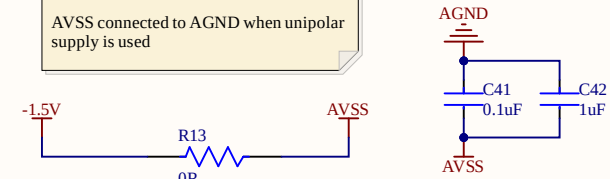


Figure 16. LM27761 Typical Application

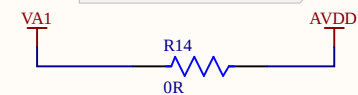
$$V_{OUT} = -1.22 \text{ V} \times (R1 + R2) / R2$$

Analog supply for the ADS can be switched on/off software controlled with the PMIC.

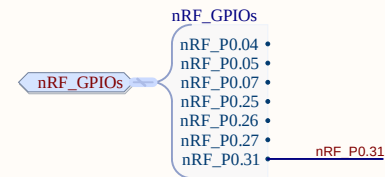
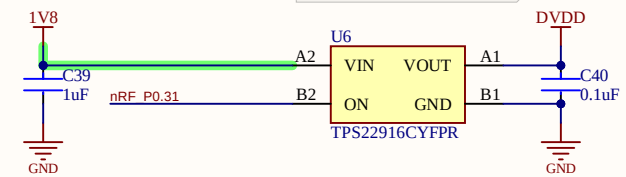
AVSS = -1.5V when bipolar supply is used
AVSS connected to AGND when unipolar supply is used



VA1 = 1.5V when bipolar supply is used
VA1 = 3.0V when unipolar supply is used



Digital supply for the ADS can be switched on/off with a load switch.



ETH

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

BioGAP ExG shield

Drawing number: 1

Rev: v1.0

Format:

Laboratory: Integrated System Laboratory

Sheet: 01_power.SchDoc

Date: 20.04.2026 15:40:20

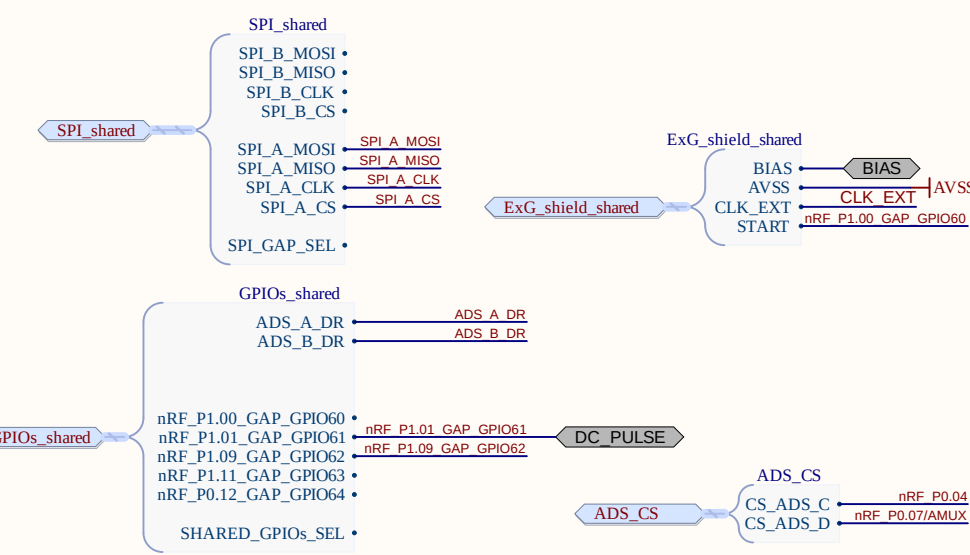
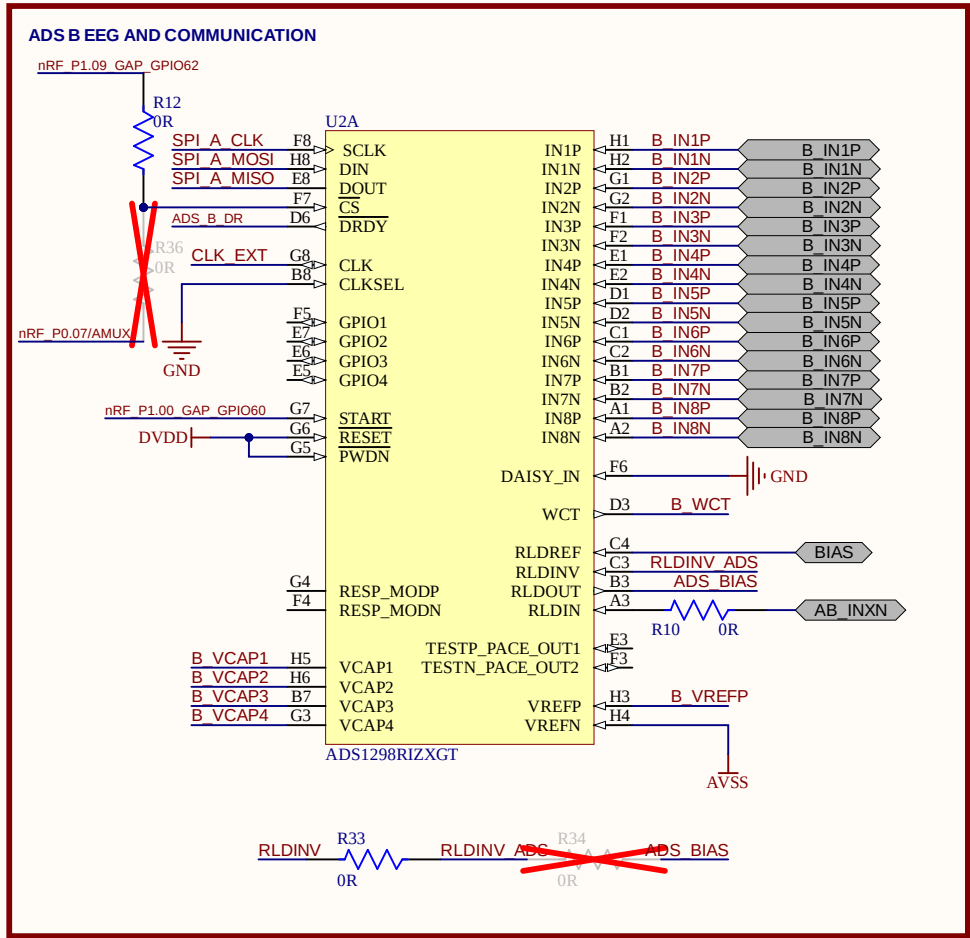
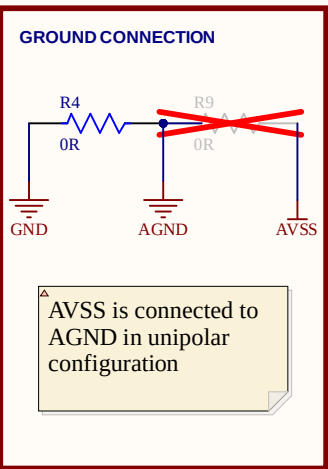
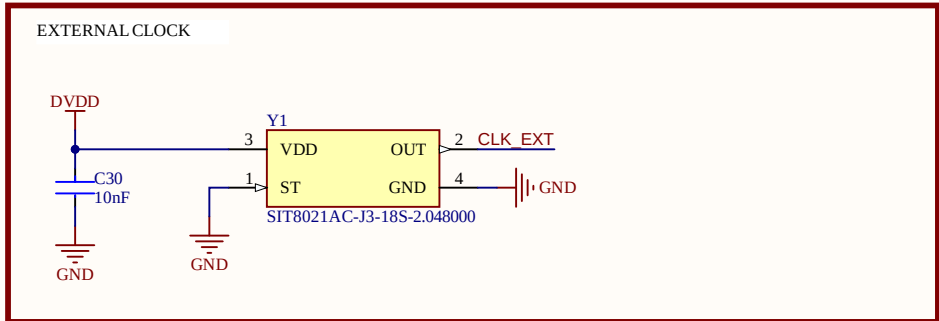
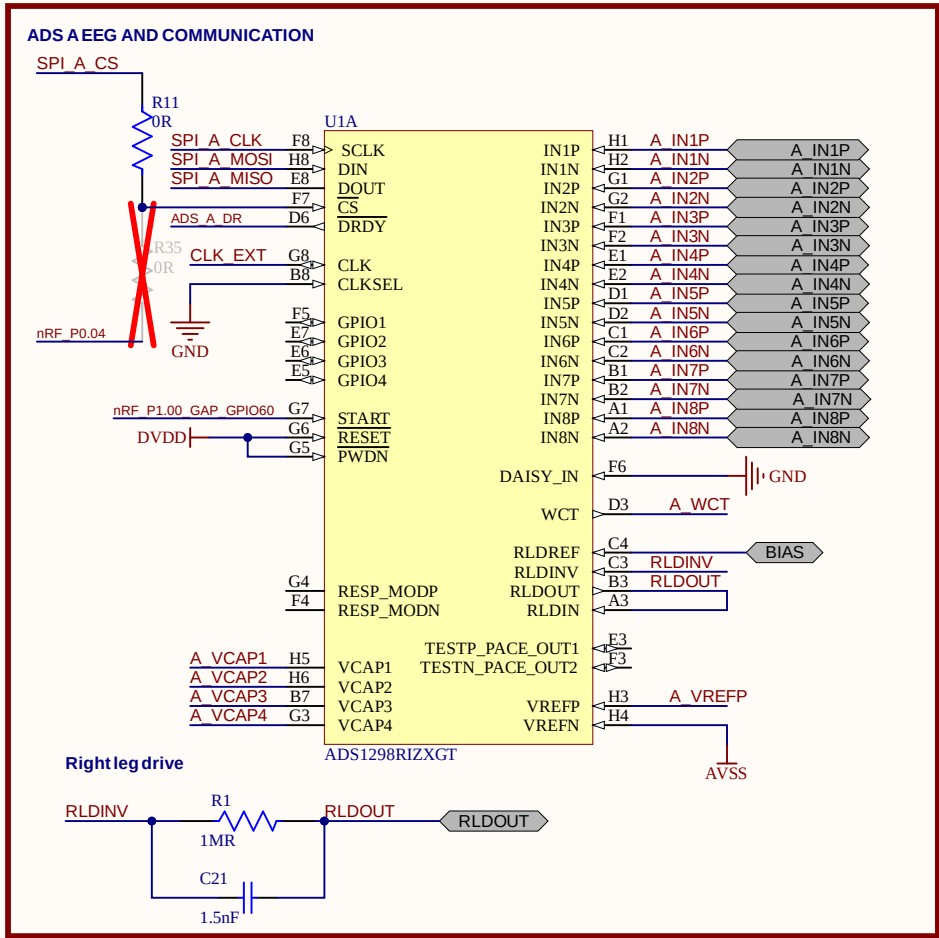
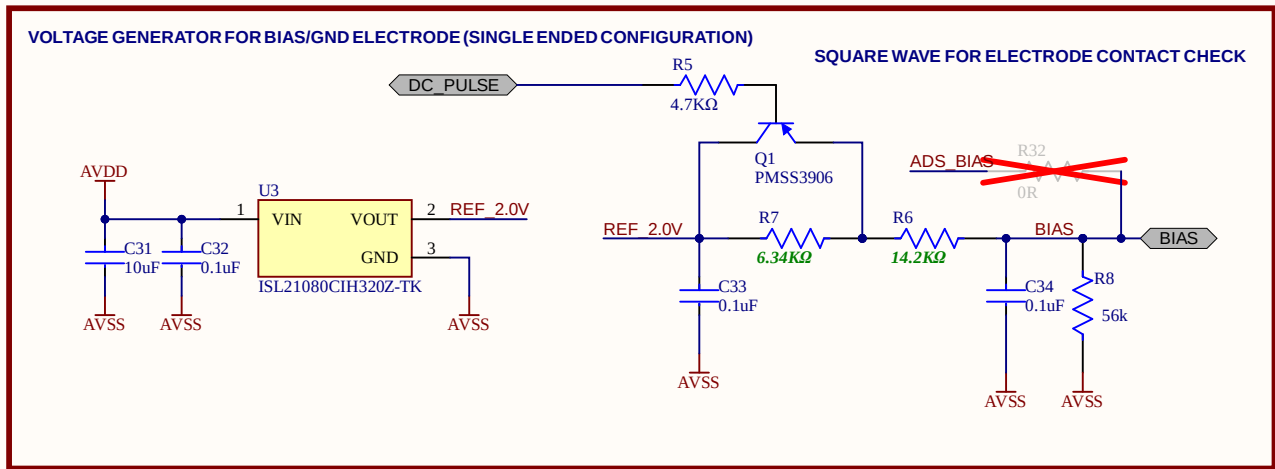
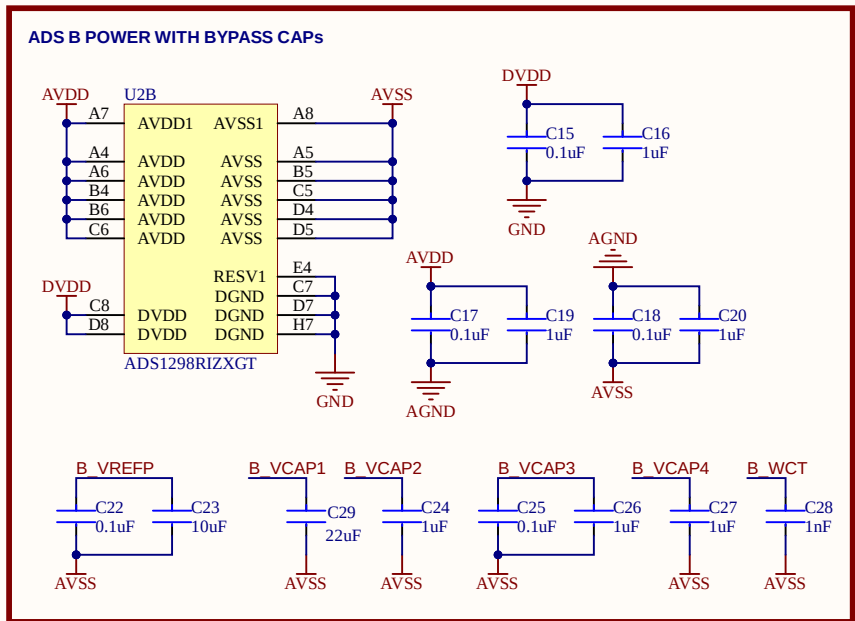
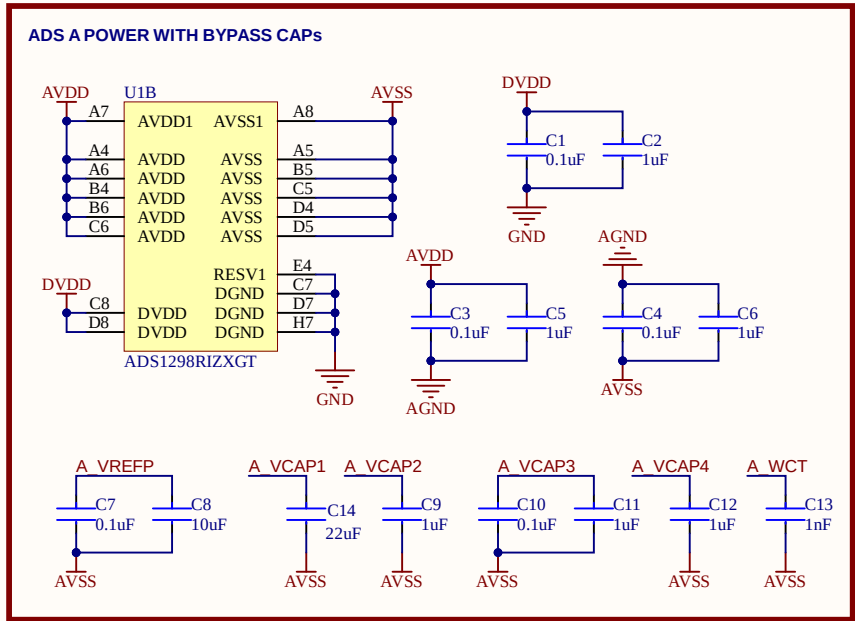
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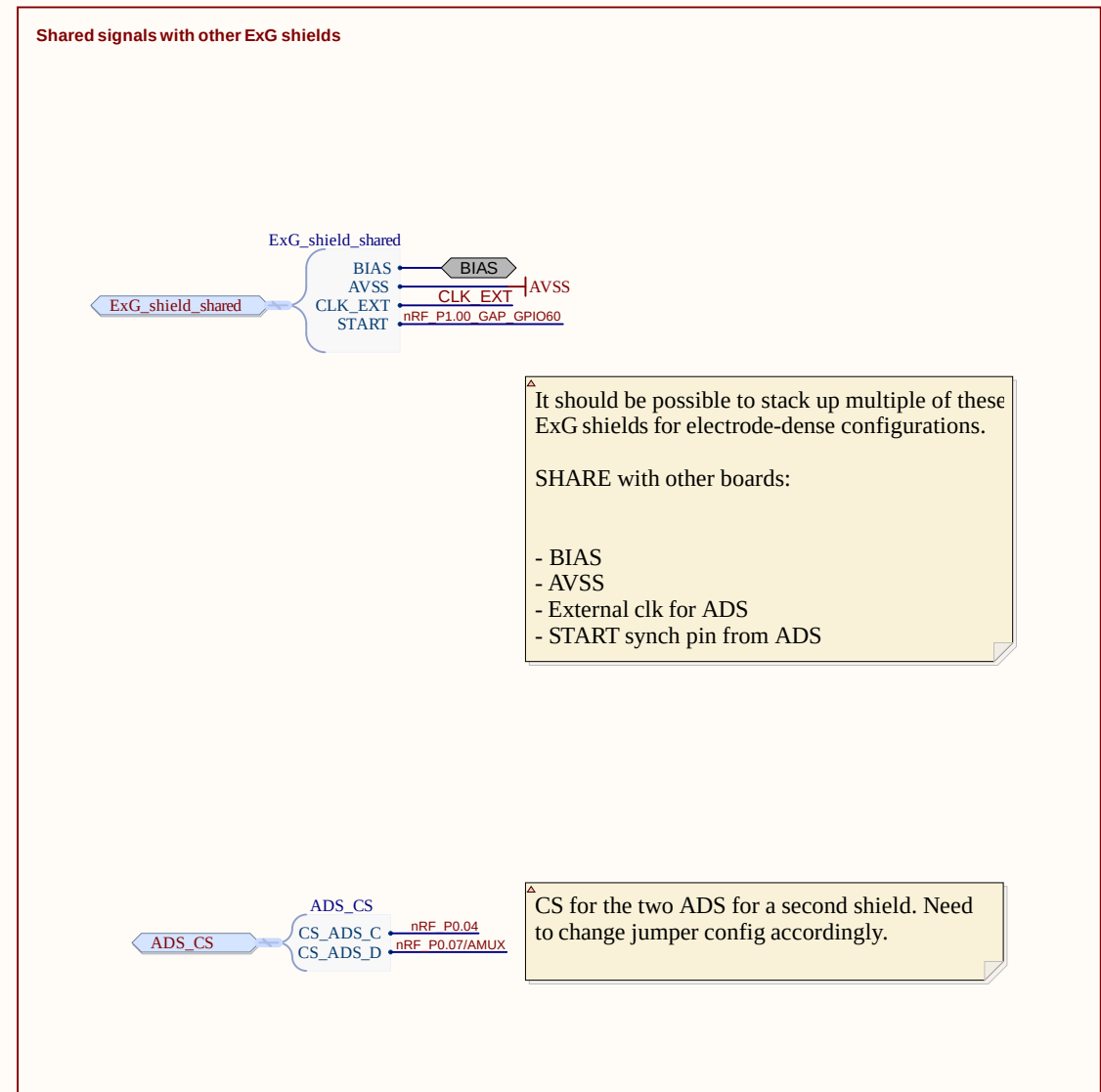
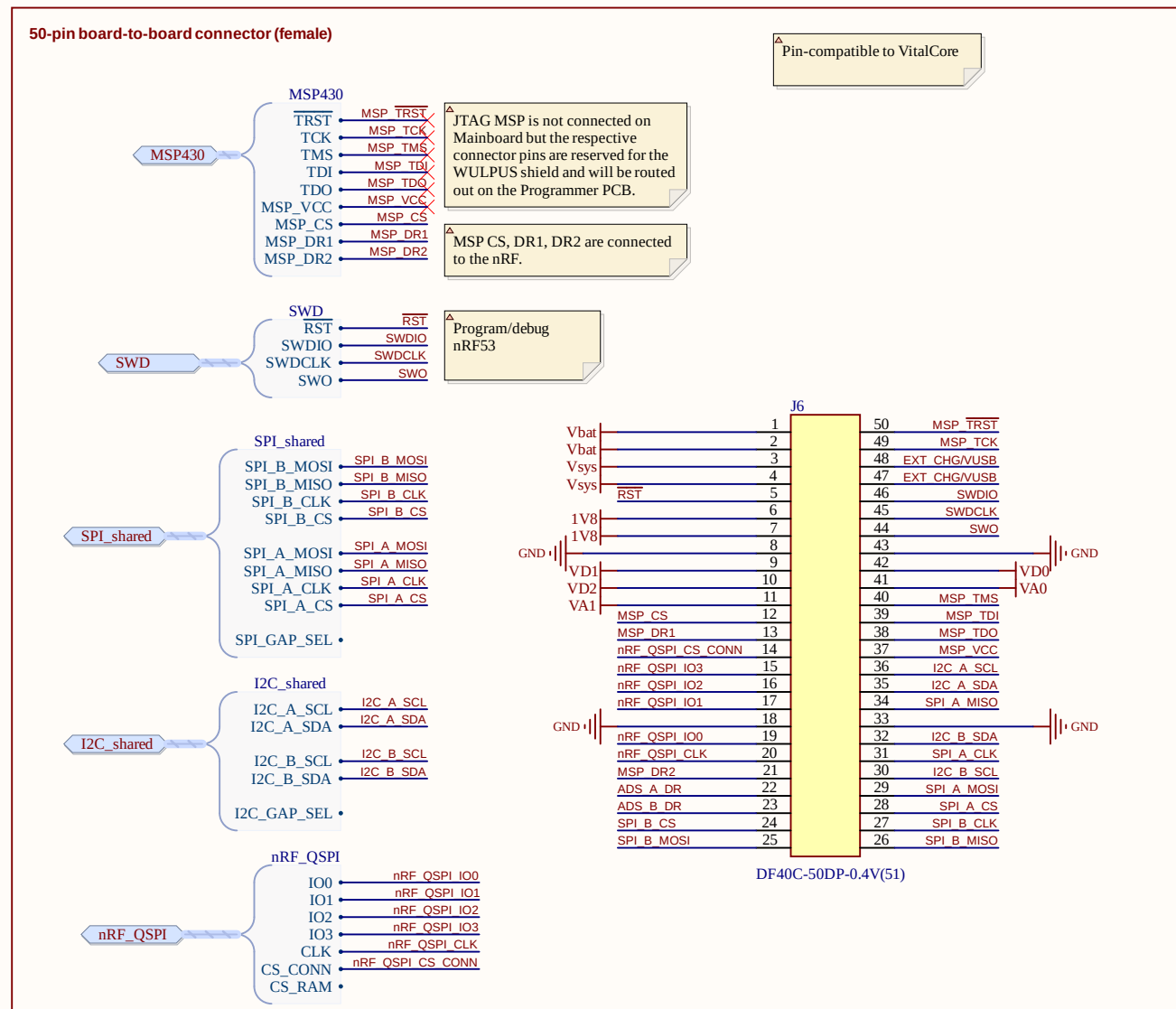
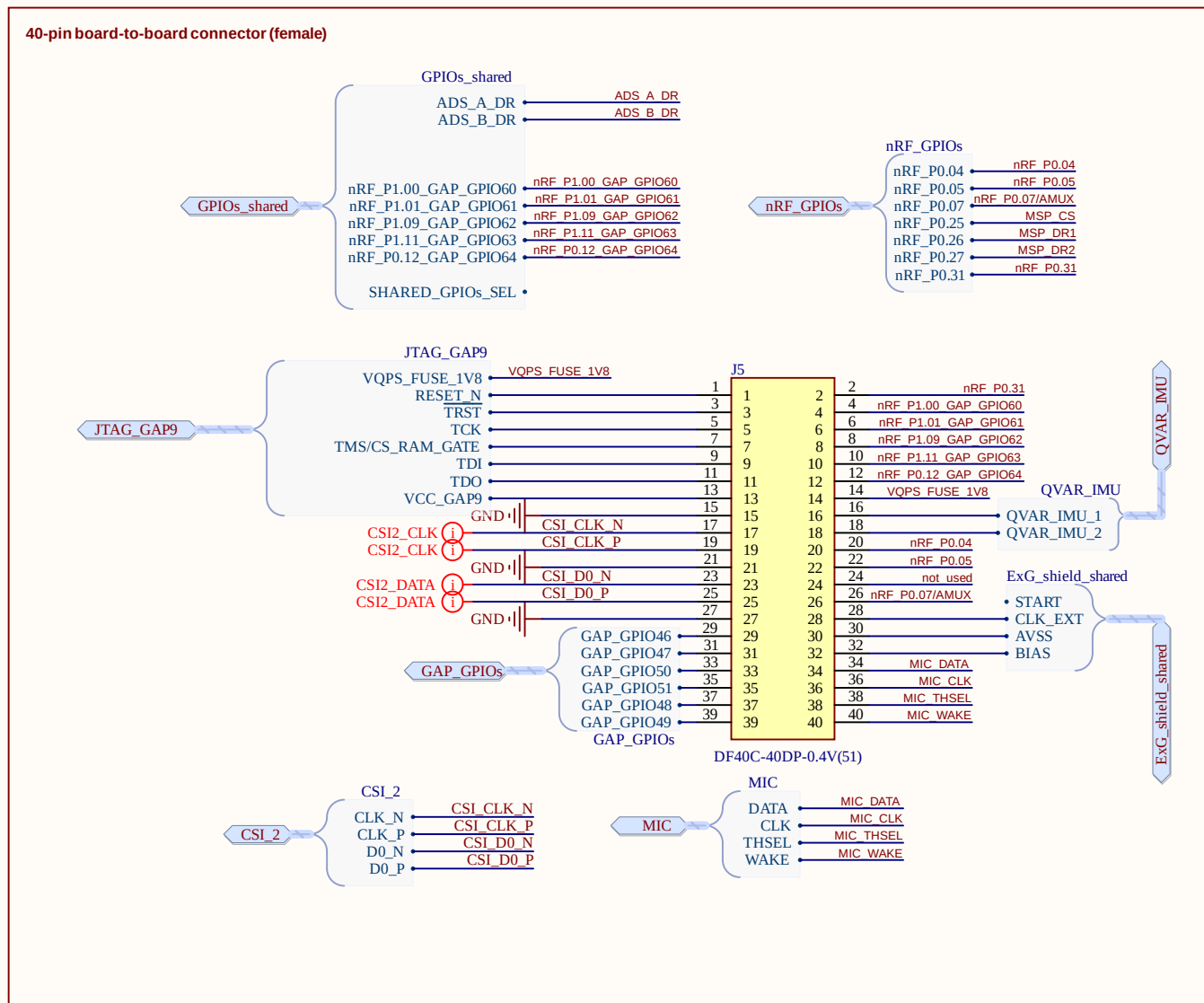
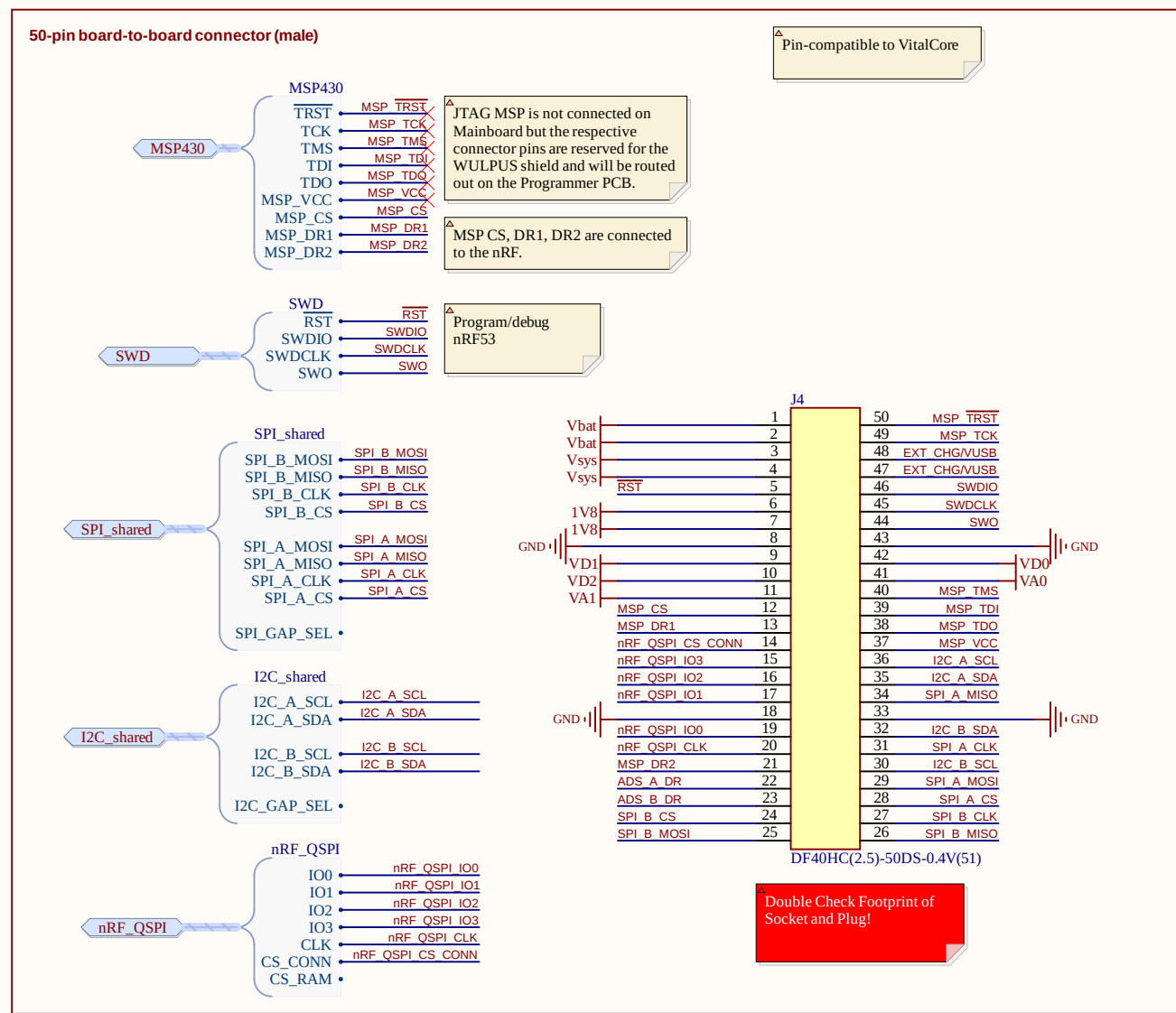
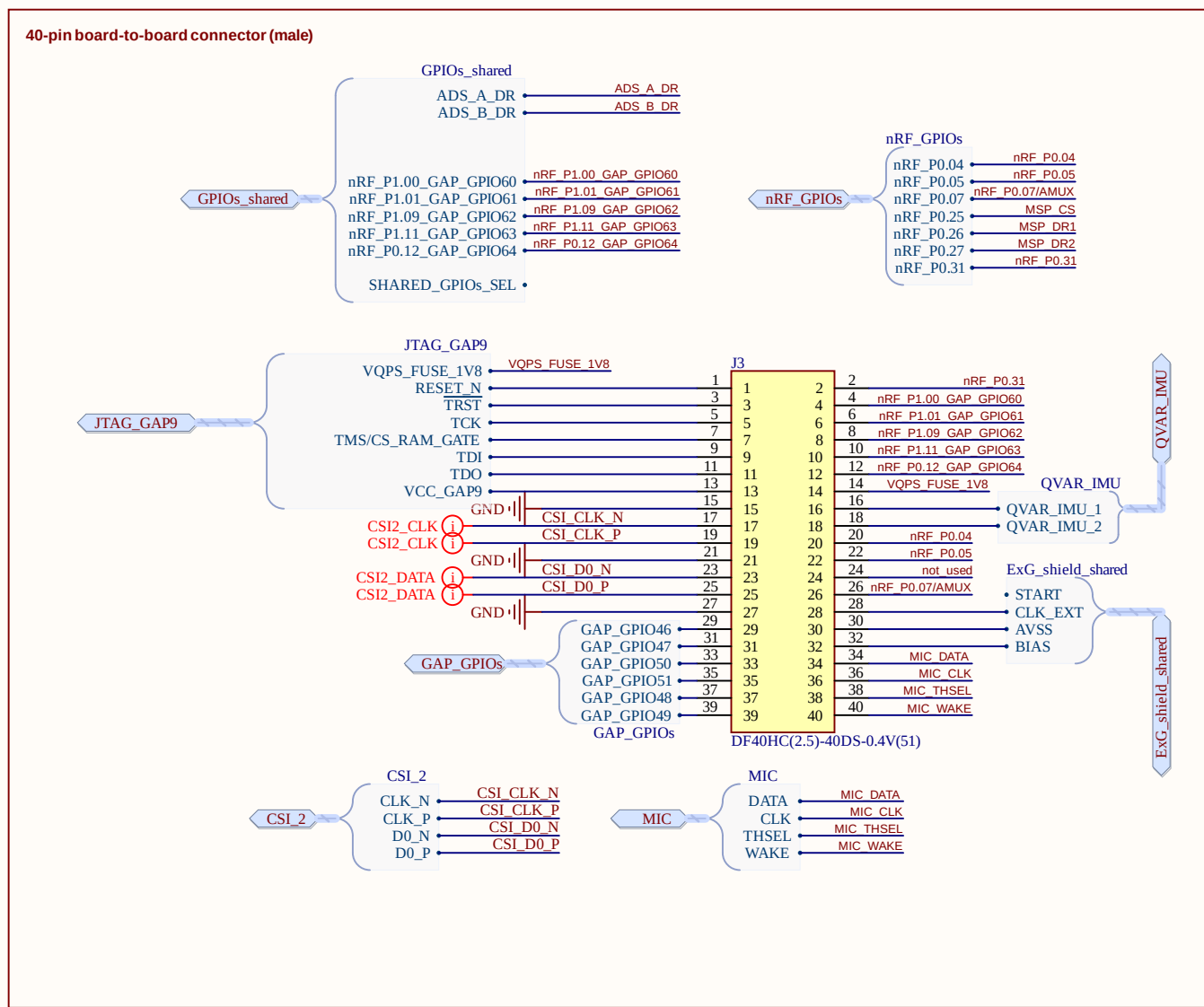
Drawn by:

Sebastian Frey

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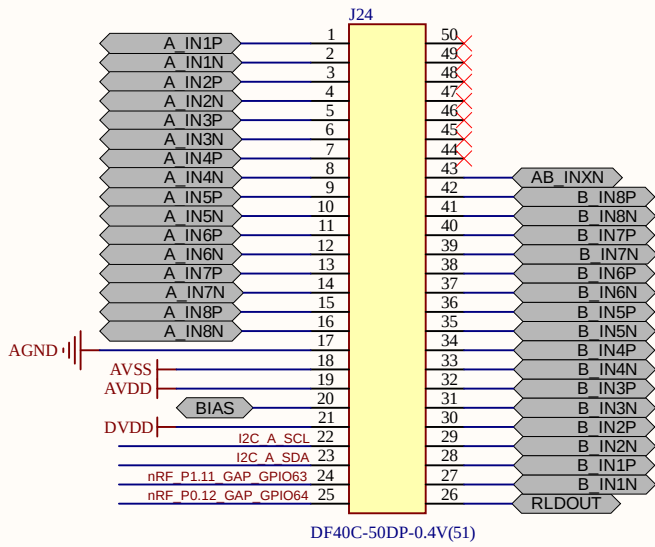
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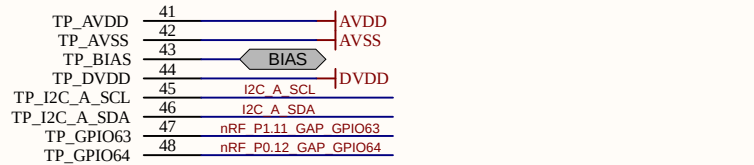
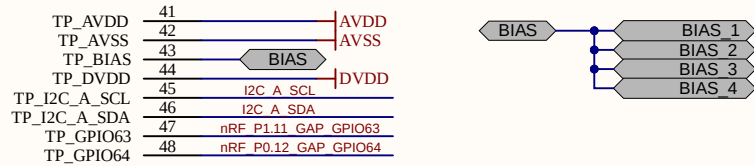
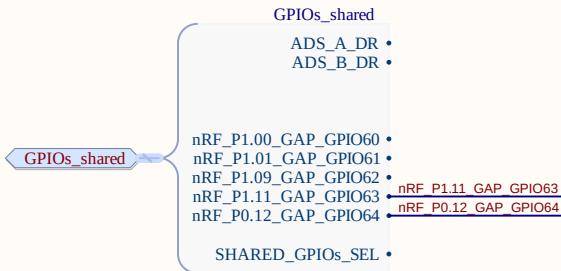
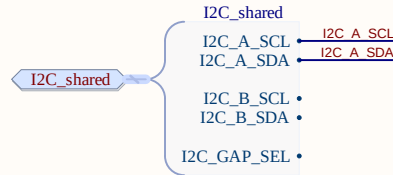
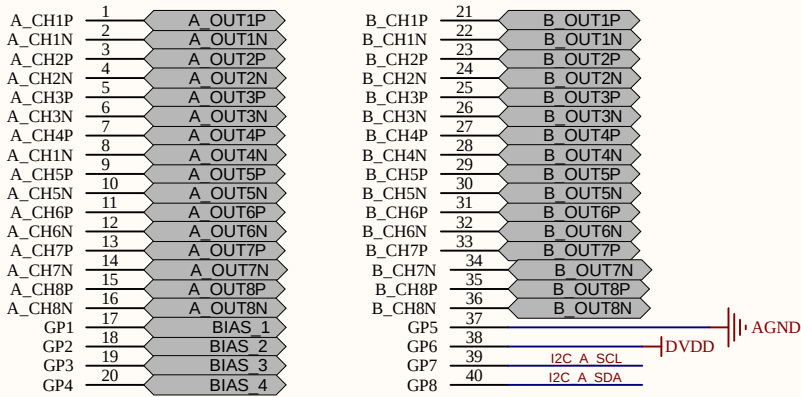
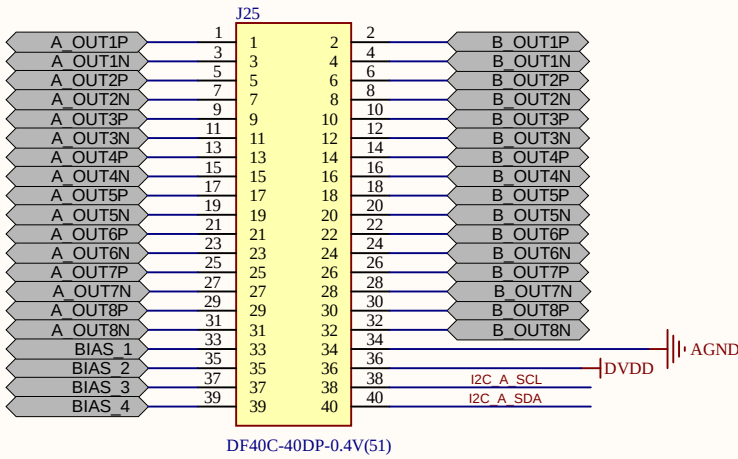


Board-to-board connector

50-pin board-to-board connector (female)



40-pin board-to-board connector (female)



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

BioGAP EMG shield

Drawing number: *

Rev: v2.0

Format:

Laboratory: *

Sheet: 03_electrode_interface_new.S

Date: 20.04.2026 15:40:21

A3

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Page * of *

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